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| **2026**<br>**MATH CP**<br>**PAPER Paper 1** |

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| Candidate Number |

# HKDSE Mathematics Compulsory Part — Paper 1 (Mock)

## PAPER Paper 1

**Question-Answer Book**

135 minutes (2.25 hours)

Total marks：105

**INSTRUCTIONS**

This paper consists of THREE sections, A(1), A(2) and B. Unless otherwise specified, all working must be clearly shown. Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures. The diagrams in this paper are not necessarily drawn to scale.

**Section A(1) (35 marks)**

1. Simplify $\frac{u^9v^{-4}}{(u^{-3}v)^2}$ and express your answer with positive indices.

(3 marks)

2. Make $h$ the subject of the formula $\frac{5h+2}{h-4} = k$ .

(3 marks)

3. ((a)) A packet of coffee beans is termed acceptable if its weight is measured as $2400$\( g\) correct to the nearest $100$\( g\) . Find the least possible weight of an acceptable packet of coffee beans.

((b)) There are $6$ acceptable packets of coffee beans. Is it possible that the total weight of these $6$ packets is $14650$\( g\) ? Explain your answer.

(3 marks)

4. ((a)) Factorize $25x^2 - 30xy + 9y^2$ .

((b)) Factorize $25x^2 - 30xy + 9y^2 + 10x - 6y$ .

(4 marks)

5. An air purifier is sold at a discount of $30\%$ on its marked price. After selling the purifier, the profit is $\$224$ and the percentage profit is $32\%$ . Find the marked price of the purifier.

(4 marks)

6. ((a)) Consider the compound inequality $4(x+2)$\( $\geq$ \)$x-1$ and $7-2x>x-8$ ……(*) . Solve (*) .

((b)) Write down the least integer satisfying (*) .

(4 marks)

7. ((a)) The coordinates of the points $A$ and $B$ are $(6, -8)$ and $(-3, 5)$ respectively. $A'$ is the reflection image of $A$ with respect to the line $y = x$ . $B$ is rotated clockwise about the origin $O$ through $90^{\circ}$ to $B'$ . Write down the coordinates of $A'$ and $B'$ .

((b)) Find the slope of $A'B'$ .

(4 marks)

8.

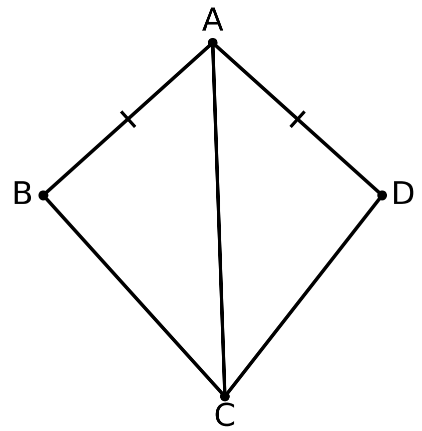


Figure 1

((a)) In Figure 1, $ABCD$ is a quadrilateral. It is given that $AB = AD$ and $\angle BAC = \angle DAC$ .
Prove that $\triangle ABC \cong \triangle ADC$ .

((b)) If $\angle ABC = 100^{\circ}$ and $\angle BCD = 96^{\circ}$ , find $\angle BAD$ .

(5 marks)

9. ((a)) The table below shows the distribution of the times taken by a group of $60$ students to finish an assignment.

Time taken (minutes): $6 - 10$ ; $11 - 15$ ; $16 - 20$ ; $21 - 25$ ; $26 - 30$

Frequency: $6$ ; $12$ ; $20$ ; $14$ ; $8$

Write down the modal class of the distribution.

((b)) Find the mean of the distribution.

((c)) A student is randomly selected from the group. Find the probability that the time taken by the selected student is less than $15.5$ minutes.

(5 marks)

**Section A(2) (35 marks)**

10. ((a)) It is given that $f(x)$ is partly constant and partly varies as $x^2$ . Suppose that $f(1) = 9$ and $f(3) = 41$ . Find $f(x)$ .

((b)) Write down the $y$-intercept of the graph of $y = f(x)$ .

((c)) Let $k$ be a real constant. Find the range of values of $k$ such that the equation $f(x) = k$ has two distinct real roots.

(7 marks)

11.

| Stem (tens) | Leaf (units) | | |
|---|---|---|---|
| 1 | 3 | 6 | 9 |
| 2 | 1 | 4 | a |
| 3 | 0 | 5 | b |
| 4 | 2 | 8 | |

Figure 2

((a)) The stem-and-leaf diagram in Figure 2 shows the distribution of the ages of the $11$ members of a society. It is given that the median and the inter-quartile range of the distribution are $26$ and $17$ respectively. Find $a$ and $b$ .

((b)) Find the mean and the standard deviation of the distribution.

((c)) A member aged $48$ now leaves the society. Find the change in the median of the distribution.

(7 marks)

12. ((a)) Let $p(x) = 2x^3 + ax^2 + bx - 18$ , where $a$ and $b$ are constants. When $p(x)$ is divided by $x^2 - 2x + 3$ , the remainder is $-5x - 15$ . Find $a$ and $b$ .

((b)) Is $x - 3$ a factor of $p(x)$ ? Explain your answer.

((c)) Someone claims that the equation $p(x) = 0$ has three distinct real roots. Do you agree? Explain your answer.

(7 marks)

13. ((a)) There are two similar solid metal right circular cones. The ratio of the volume of the smaller cone to the volume of the larger cone is $64 : 125$ . The base radius and the height of the larger cone are $15$\( cm\) and $20$\( cm\) respectively. Express, in terms of \($\pi$\) , the volume of the smaller cone.

((b)) The two cones are melted and recast into a solid right circular cylinder of base radius $6$\( cm\) . Find the height of the cylinder.

((c)) Someone claims that the curved surface area of the cylinder is greater than the curved surface area of the larger cone. Do you agree? Explain your answer.

(7 marks)

14. ((a)) The equation of the circle $C$ is $x^2 + y^2 - 12x - 20y + 72 = 0$ . Denote the centre of $C$ by $G$ . Find the coordinates of $G$ and the radius of $C$ .

((b)) The coordinates of the point $H$ are $(21, 10)$ . Find the distance between $G$ and $H$ .

((c)(i)) Let $P$ be a moving point on $C$ . When the area of $\triangle GHP$ is the greatest, describe the geometric relationship between $GH$ and $GP$ .

((c)(ii)) Find the perimeter of $\triangle GHP$ in this case.

(7 marks)

## Section B (35 marks)

15. ((a)) In a box, there are **7** red balls, **5** blue balls and **3** green balls. **4** balls are randomly drawn from the box at the same time. Find the probability that exactly **2** red balls are drawn.

((b)) Find the probability that at least **1** red ball and at least **1** green ball are drawn.

(6 marks)

16. ((a)) Let $f(x) = 2x^2 - 8kx + 10k^2 + 24k + 7$ , where $k$ is a real constant. Using the method of completing the square, express, in terms of $k$ , the coordinates of the vertex of the graph of $y = f(x)$ .

((b)) Find the least value of the $y$-coordinate of the vertex and the corresponding value of $k$ .

((c)) Someone claims that for any real constant $k$ , the vertex of the graph of $y = f(x)$ lies on the curve \(Γ\) : $y =$ \($\frac{1}{2}$\)$x^2 + 12x + 7$ . Do you agree? Explain your answer.

(6 marks)

17. ((a)) Let $A(n)$ be the $n$th term of an arithmetic sequence. It is given that $A(2)=11$ and $A(7)=41$ . Find $A(n)$ .

((b)) Find $A(1)+A(2)+$ \($\cdots$\) $+A(40)$ .

((c)) Let $G(n)$ be the $n$th term of a geometric sequence with first term $A(3)$ and common ratio $\mathbf{2}$ . Find the least value of $n$ such that

$G(1)+G(2)+$ \($\cdots$\) $+G(n)>A(1)+A(2)+$ \($\cdots$\) $+A(40)$ .

(6 marks)

18.

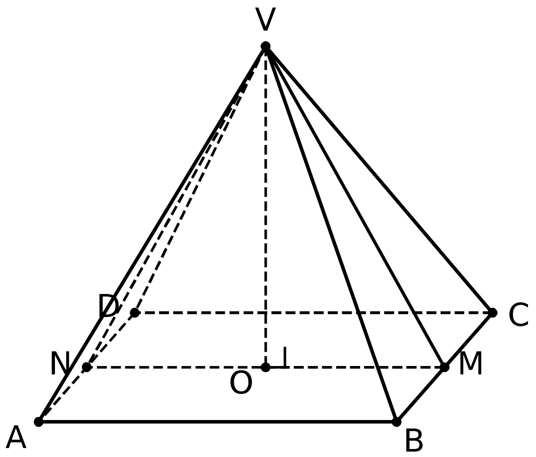


Figure 3

((a)) Figure 3 shows a right pyramid $VABCD$ with a square base $ABCD$ of side $16\ \text{cm}$ . The height of the pyramid is $15\ \text{cm}$ . Let $M$ be the mid-point of $BC$ and $O$ be the centre of $ABCD$ . Find $VM$ .

((b)) Find the angle between the plane $VBC$ and the base $ABCD$ .

((c)) Let $N$ be the mid-point of $AD$ . A student claims that the angle between the plane $VBC$ and the plane $VAD$ is less than $60^{\circ}$ . Do you agree? Explain your answer.

(8 marks)

19. ((a)) The equation of the circle $C$ is $x^2 + y^2 - 10x - 24y = 0$ . Denote the centre of $C$ by $G$ . Find the coordinates of $G$ and the radius of $C$ .

((b)) Let $L : 5x + 12y - 338 = 0$ . Prove that $L$ is a tangent to $C$ , and find the coordinates of the point of contact $T$ .

((c)) Denote the origin by $O$ . Someone claims that $OT$ is a diameter of $C$ . Do you agree? Explain your answer.

((d)) Let $P$ be a moving point on $C$ such that $P$ is different from $O$ and $T$ . Denote the area of \($\triangle OPT$\) by $S$ . Find the greatest value of $S$ .

(9 marks)

**END OF PAPER**